Nahid Kalantaryardebily

Blacksburg-USA NahidKalantary@vt.edu nahid-Kalantaryardebily Google Scholar

Selected Highlights

- 9 peer-reviewed journal publications, including 3 first/co-first-author articles in IEEE Transactions on Medical Robotics and Bionics, Journal of Neuroscience Methods, and Extreme Mechanics Letters.
- Co-inventor on 1 provisional patent and Virginia Tech technology disclosure
- Completed 15 peer-review assignments for 8 manuscripts across 3 journals and 2 IEEE conferences
- Contributed to >$40,000 in research funding
- Recipient of the Joseph Frank Hunkler Memorial Scholarship (includes a $35,000 annual fellowship and full in-state tuition support) and the Daniel Frederick Scholarship.

Research Interest

Machine Learning for Robotic Control and Perception; Robotics and Automation; Tactile Sensing and Haptics; Human–Robot Interaction; Neuroengineering and Sensorimotor Neural Interfaces; Wearable Technology.

Education

Ph.D.	**Engineering Mechanics** (GPA: 4/4) - Ph.D. Candidate Virginia Polytechnic Institute & State University	Blacksburg, USA 2022 – Present
M.S.	**Engineering Mechanics** (GPA: 4/4) Virginia Polytechnic Institute & State University	Blacksburg, USA 2022 – 2025
M.S.	**Automation and Control Engineering** (GPA: 110/110) Polytechnic University of Milan Thesis: Design and Development of a Hardware-in-the-Loop Test Platform for Powertrain	Milan, Italy 2016 – 2018
M.S.	**Flight Mechanics** (GPA: 18.6/20) Amirkabir University of Technology Thesis: Integrated Guidance and Control of UAVs Using Sliding Mode Control and Constrained Curves	Tehran, Iran 2009 – 2011
B.S.	**Electrical Engineering - Control** (GPA: 17.2/20) Amirkabir University of Technology	Tehran, Iran 2005 – 2013
B.S.	**Aerospace Engineering** (GPA: 17.8/20) Amirkabir University of Technology	Tehran, Iran 2005 – 2009

Awards & Honors

- **Joseph Frank Hunkler Memorial Scholarship**, Graduate School, Virginia Tech — competitive fellowship awarded for research promise and academic achievement; includes a $35,000 annual stipend and full in-state tuition support. 2026 - 2027
- **Daniel Frederick Scholarship**, Department of Mechanical Engineering, Virginia Tech — merit-based award for academic excellence, research creativity, and strong doctoral exam performance. 2025 - 2026
- **Direct Admission and Full Scholarship to Master's Program in Flight Mechanics**, Amirkabir University of Technology — competitive honor granted without an entrance exam for top-ranked students. 2008 - 2009
- **Merit-Based Scholarship for a Second Bachelor's Degree**, Amirkabir University of Technology — competitive scholarship awarded for strong academic performance. 2007 - 2008

Grants

- Title: Impact of Stroke on Peripheral Nerve Signaling 2026 - 2028
  Funder: National Institutes of Health
  PI: Netta Gurari
  Role: Graduate Research Assistant
  Contribution: Involved with grant writing and preliminary data collection
  Amount Awarded: $454,893

- Title: Open–Access Materials for Educating Students on the Lived Brain Injury Experience 2026 - 2027
  Funder: Dana Foundation NextGen
  PI: Netta Gurari
  Role: Graduate Research Assistant
  Contribution: Involved with course design and implementation, and documentation
  Amount Awarded: $75,000
- Title: Impact of Movement Following Stroke on Perceiving Physical Stimuli during Bimanual Tasks 2025 - 2026
  Funder: 4-VA Collaborative Research Grant
  PI: Netta Gurari
  Role: Graduate Research Assistant
  Contribution: Involved with grant writing, implementation, and data collection
  Amount Awarded: $30,000
- Title: Impact of Movement on Perceiving Touch Stimuli during Tasks that use Both Arms following Stroke 2025 - 2026
  Funder: Radford University SEED Grant Program
  PI: Arco Paul
  Role: Graduate Research Assistant
  Contribution: Involved with grant writing, implementation, and data collection
  Amount Awarded: $10,000

Journal Publications

———————————————————- First/Co-First Author Journal Articles ———————————————————-

- **Kalantaryardebily N.**, Feldbush A. C., Kahak A., Li S., Gurari N. “Novel Compact Tactile Stimulator with Sensing: Application for Individuals with a Brain Injury and MRI” *IEEE Transactions on Medical Robotics and Bionics* (2026).
- Kahak A.*, Ahmed M. S.*, **Kalantaryardebily N.***, Kulkarni H., Gurari N., Shahab S., Li S. “High-Precision Fluidic Kirigami Metasurface for Ultrasonic Holographic Lensing and Haptic Interfacing” *Extreme Mechanics Letters*, 81, 102424 (2025). *- Authors marked with * contributed equally*
- **Kalantaryardebily N.***, Feldbush A. C.*, Faubion-Trejo R., Lisinski J., Reddy N. A., Bright M. G., LaConte S. M., Gurari N. “Development and Testing of an MR-Compatible Tactile Stimulator System: Application for Individuals with a Brain Injury” *Journal of Neuroscience Methods*, 424, 110583 (2025). *- Authors marked with * contributed equally*

————————————————————— Co-Author Journal Articles —————————————————————

- Tirrell E. M., **Kalantaryardebily N.**, Hocker J., Bowles C., Parcetich K., Gurari N. “Perceiving Auditory Stimuli Impacts Conscious Perception of Electrotactile Stimuli in Older Adults” *Journal of Neurophysiology* (2026).
- Tirrell E. M., **Kalantaryardebily N.**, Hocker J., Bowles C., Parcetich K., Gurari N. "Impact of an Auditory Cognitive Load on Consciously Perceiving Electrotactile Stimuli in Young Adults” *Behavioral Brain Research*, 496, 115841 (2026).
- Feldbush A. C., **Kalantaryardebily N.**, Reddy N. A., Soldate J., Faubion-Trejo R., Lisinski J., Bright M. G., Gurari N., LaConte S. M. “Tactile Pressure Evokes a Biphasic BOLD Response in Ipsilateral Primary Somatosensory Cortex” *Imaging Neuroscience* (2026).
- Tirrell E. M., **Kalantaryardebily N.**, Feldbush A. C., Sydnor L. C., Grubb C., Parcetich K., Gurari N. “Considerations for Tactile Perceptual Assessments: Impact of Arm Dominance, Nerve, Location, and Sex in Young and Older Adults” *Experimental Brain Research*, 243, 92 (2025).
- Paul A. P., Nayak K., Sydnor L., **Kalantaryardebily N.**, Parcetich K. M., Miner D. G., Wafford Q. E., Sullivan J. E., Gurari N. “A Scoping Review on Examination Approaches for Identifying Tactile Deficits at the Upper Extremity in Individuals with Stroke” *Journal of Neuroengineering and Rehabilitation*, 21, 99 (2024).
- Hamedi B., **Kalantaryardebily N.**, Alipanahi A., Taheri S. “Estimating Tire Forces using MLP Neural Network and LM Algorithm: A Comparative Study” *Universal Journal of Mechanical Engineering*, 11(3), 64–81 (2023).

—————————————————— Research Manuscripts in Preparation ——————————————————

- **Kalantaryardebily N.**, Mahany S., Parcetich K., Gurari N. “Development and Validation of a Somatosensory Assessment System for Investigating the Impact of Voluntary Movement on Tactile Perception” (In preparation)
- **Kalantaryardebily N.**, Ahrens M., Yau J., Paul A., Gurari N. “Impact of Arm Activation on Detecting Bilateral Stimulation” (In preparation)
- **Kalantaryardebily N.**, Parcetich K., Nape S., Sulzer J., Pitterson N., Gurari N. “Immersive Neurorehabilitation Education Improves Competencies Relevant to Human-Centered Biomedical Engineering Practice” (In preparation)

Intellectual Property

- **Kalantaryardebily N.**, Gurari N., Kahak A., Li S. "Compact MRI-Safe Tactile Stimulator with Precise Force Control" Provisional Patent Application, Virginia Tech Intellectual Property (VTIP), Docket No. 222207-8210 (2025). Technology Available for Licensing

Presentations

—————————————————— **Peer-Reviewed Conference Publications** ——————————————————-

- Rekabi-Ban F., Saadat M., **Kalantaryardebily N.**, Pan H., Stefanec M., Fedotoff L. A., Krajnik T., Arvin F. "Active Vibration Reduction for the RoboRoyale Autonomous Robotic Observation Mechanism" *2024 IEEE Conference on Control Technology and Applications (CCTA)*, Newcastle upon Tyne, United Kingdom (2024).
- Basaeri H., Yousefi-Koma A., **Kalantaryardebily N.** "Design and Modeling of a Ducted Fan Manned Aerial Vehicle" *21st Annual International Conference on Mechanical Engineering – ISME2013*, Tehran, Iran (2013).
- **Kalantaryardebily N.**, Mortazavi M., Babai A. "Integrated Guidance and Control of UAVs Using Sliding Mode Control and Constrained Curves" *11th Iranian Aerospace Conference*, Tehran, Iran (2012). In Persian.

—————————————————————— **Hands-On Demonstrations** ——————————————————————

- Rathore S., Kim A., Segal J., Hollar R., Smith N., Rios J., **Kalantaryardebily N.***, Feldbush A. C., Todd J., Parcetich K., Gurari N. "MR-Compatible Tactile Stimulator for Individuals with Stroke" World Haptics Conference (WHC), Delft, Netherlands (2023) ***Author marked with * denotes the presenter***

——————————————————— **Peer-Reviewed Abstract Publications** ———————————————————-

- **Kalantaryardebily N.**, Ahrens M., Yau J., Paul A., Gurari N. "Sensorimotor Gating of Tactile Detection Scales with Motor Activation During Unilateral and Bilateral Elbow Flexion." (Accepted)
- Gurari N., **Kalantaryardebily N.**, Nape S., Pitterson N., Sulzer J., Parcetich K. "Immersive Neurorehabilitation Education Improves Self-Reported Competencies Relevant to Human-Centered Biomedical Engineering Practice." (Accepted)
- **Kalantaryardebily N.**, Kahak A., Mahaney S., Soldate J., Li S., Gurari N. "Development and Preliminary Testing of a Compact Stimulator for Objective Tactile Assessment." Poster presented at American Society of Neurorehabilitation (ASNR), Long Beach, California (2026)
- Feldbush A., **Kalantaryardebily N.**, Reddy N., Faubion-Trejo R., Soldate J., Lisinski J., Bright M. G., Gurari N., LaConte S. M. "Tactile Pressure Evokes Biphasic BOLD Response in Ipsilateral Primary Somatosensory Cortex." Poster presented at Organization for Human Brain Mapping (OHBM), Brisbane, Australia (2025)
- Nayak K. S., Sydnor L. C., Parcetich K., Paul A., Miner D. G., **Kalantaryardebily N.**, Wafford Q. E., Sullivan J. E., Gurari N. "Assessments for Identifying Tactile Deficits at the Upper Extremity of Individuals with Stroke: Preliminary Results of a Scoping Review," Poster presented at American Society for Neurorehabilitation, San Antonio, Texas, USA (2024)

——————————————————– **Non-Peer-Reviewed Abstract Publications** ——————————————————

- **Kalantaryardebily N.**, Gurari N. "Development and Characterization of a Novel MR-Compatible Tactile Stimulator with Force Estimation." Poster presented at Virginia Tech Graduate Research Symposium, Blacksburg, USA (2025)
- Elahimehr N., **Kalantaryardebily N.**, Gurari N. "Measuring Peripheral Nerve Sensory Function Using Tactile Stimulation." Poster presented at Virginia Tech Research Symposium, Blacksburg, USA (2025)
- Vijay R., **Kalantaryardebily N.**, Feldbush A., LaConte S. M., Gurari N. "Characterizing a Novel MR-Compatible Force Stimulator". Poster presented at Virginia Tech, Fralin Biomedical Research Institute, Summer Research Programs Symposium, Roanoke, USA (2024)

Peer-Review Activities

- IEEE Transactions on Haptics (IEEE) - 8 reviews
- Data in Brief (Elsevier) - 2 reviews
- Discover Neuroscience (Springer Nature) - 3 reviews
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) - 1 review
- IEEE Conference on Control Technology and Applications (CCTA) - 1 review

Teaching Experience

Teaching Assistant, Virginia Tech, Undergraduate level, Neurorehabilitation Perspectives Spring 2026

- Assisted in organizing the course material, logistics and immersion experience.

Teaching Assistant, Virginia Tech, Graduate level, ME Graduate Seminar Fall 2025

- Assisted in organizing and advertising Mechanical Engineering seminar sessions.

Teaching Assistant, Virginia Tech, Undergraduate level, Mechanics of Deformable Bodies Fall 2024

- Assisted in grading homework and course projects.

Teaching Assistant, Virginia Tech, Undergraduate level, Statics & Structures Spring 2023, Fall 2022

- Assisted in grading midterm exams and held office hours to support students with problem-solving and homework.

Teaching Assistant, Amirkabir University of Technology, Undergraduate level, Dynamics I 2010 - 2013, 2 Semesters

- Developed and delivered problem-solving lectures for undergraduate courses in Dynamics I, grading homework and Midterm exam.

Teaching Assistant, Amirkabir University of Technology, Undergraduate level, Dynamics II 2010 - 2013, 2 Semesters

- Developed and delivered problem-solving lectures for undergraduate courses in Dynamics II, grading homework and Midterm exam.

Teaching Assistant, Amirkabir University of Technology, Undergraduate level, Vibration 2009 - 2013, 3 Semesters

- Developed and delivered problem-solving lectures for undergraduate courses in Vibration, grading homework and Midterm exam.

Mentoring Experience

Masters Student, Biomedical Engineering — Scotty Mahaney	2025 – Present
Undergrad Student, Neuroscience — Nithya Ramani	2025 – Present
Undergrad and Masters Student, Biomedical Engineering — Rishit Vijay	2023 – 2026
Senior Design Team (Biomedical Engineering) — Smart Goggle Project	2025 – 2026
Undergrad Student, Biomedical Engineering — Saikrishna Ramasamy	Feb – May 2026
Undergrad Student, Biomedical Engineering — Suhana Bhargava	2025 – 2026
Undergrad Student, Biomedical Engineering — Adharsh Sivaganesan	Sept – Dec 2025
Senior Design Team (Mechanical Engineering) — Tactile Stimulator Project	2023 – 2024
Undergrad Student, Biomedical Engineering — Katie Minutillo	2023 – 2024
Undergrad Student, Biomedical Engineering — Anwitha Sanivarapu	2023 – 2024

Work Experience

Virginia Tech — Robotics and Sensorimotor Control Laboratory

Blacksburg, USA
2023 – Present

Job title: Ph.D. Researcher – Robotics Systems & Software Engineer

- Designed and developed a compact, pneumatically controlled, MRI-compatible soft haptic device incorporating a Kirigami-based structure, fiber-optic displacement sensing, and pressure sensing, enabling precise force delivery from 0–3.8 N with <0.01 N force modulation.
- Developed real-time force estimation algorithms for the MRI-compatible tactile stimulator using physics-based modeling and neural networks, achieving force estimation errors of 0.087 ± 0.174 N (physical model) and 0.060 ± 0.031 N (neural network) at a 3.57 N applied force level.
- Designed and built experimental test platforms, calibration protocols, and integrated data acquisition systems combining force, displacement, and pressure sensing for device parameter identification and validation of physics-based and neural-network force estimation models.
- Developed GUI-based software for real-time experimental control, data acquisition, visualization, and analysis in tactile perception studies.
- Developed integrated software and experimental systems using Quanser and NI DAQ platforms for 1 kHz real-time measurement and quantitative assessment of peripheral nerve responses to tactile and electrical stimuli.
- Evaluated the reliability and repeatability of median and ulnar nerve excitability and conduction measurements under tactile and electrical stimulation.
- Applied threshold–tracking techniques to quantitatively evaluate peripheral nerve excitability in

individuals with stroke.

- Designed and developed integrated mechatronic experimental platforms for movement-constrained sensorimotor testing, incorporating force/torque sensing and tactile/electrical stimulation to evaluate changes in human sensory perception during movement.
- Conducted human-subject studies with individuals with brain injury and healthy adults to quantitatively assess sensory perception and sensorimotor function.
- Lead and mentor a student team in developing assistive goggle technology for individuals with brain injury and vision impairments.

SEGULA Technologies — General Motors Projects

Turin, Italy
2019 – 2022

Job title: **Software Integration Engineer**

- Created an active thermal management model and updated sensor and actuator models for diesel engines within a real-time Software-in-the-Loop (SiL) framework.
- Developed automated validation tools for Software-in-the-Loop (SiL) testing and release verification of powertrain software.
- Debugged and resolved software integration issues across embedded vehicle control systems, including CAN-based communication, signal validation, and system-level troubleshooting.
- Enhanced the integrated real-time operating system for diesel engine SiL software to ensure compatibility with the Functional Mock-up Interface (FMI) standard.

TUMCREATE Research Center

Singapore, Singapore
03/2018 – 09/2018

Job title: **Vehicle Dynamics & HiL Simulation Engineer**

- Designed, prepared, and executed the Hardware-in-the-Loop (HiL) platform using dSPACE.
- Developed, implemented, and validated a 14-degree-of-freedom real-time simulation.
- Configured TCP/IP Ethernet communications between testbench and dSPACE.
- Designed ControlDesk human–machine interface for real-time data acquisition and testing.
- Created urban environment visualization in Motion-Desk software for a driving simulator.

Tehran University — CAST Laboratory

Tehran, Iran
2011 – 2015

Job title: **Simulation & Control Engineer**

- Conducted comprehensive modeling and design verification for a manned ducted fan aerial vehicle (Jetpack).
- Formulated and implemented simulation-driven gain scheduling PID controller design.
- Developed dynamic models, simulations, and control algorithms for underwater vehicles and quadrotors.

Iran Space Research Center

Tehran, Iran
2010 – 2011

Job title: **Satellite Dynamics & Control Systems Engineer**

- Developed dynamic models, simulations, and feedback-linearization controllers for CubeSat attitude determination and control systems.
- Validated attitude determination and control algorithms using SiL and HiL testing.
- Designed satellite test facilities, including a sun simulator and mass-balancing system for a 3-degree-of-freedom air-bearing test platform.

Amirkabir University of Technology microsatellite – AUTSAT1

Tehran, Iran
06/2008 – 09/2008

Job title: **Internship**

- Identified methods, assumptions, and procedures for sun sensor and magnetometer calibration.

Media Coverage

Media Coverage of Me

- **Kalantaryardebily is chosen for Hunkler Memorial Scholarship** 07/2026
  Virginia Tech News
- **Nahid Kalantaryardebily Awarded Daniel Frederick Scholarship** 12/2025
  Virginia Tech News

Media Coverage of My Work

- **Training Scientists for a Changing World: Seeding New Models of Scientific Training** 06/2026
  Dana Foundation
- **"Lived Experience" in Neurorehabilitation: Contribution to Course Development** 05/2026
  Virginia Tech News
- **Swimming Goggle Project for Individuals With Brain Injury (Project Co-Initiator)** 05/2026
  Virginia Tech News
- **Researching How Stroke Affects Sense of Touch** 05/2024
  Virginia Tech News
- **Experiential Learning in Brain Injury Rehabilitation** 03/2024
  Virginia Tech News

Computer Skills

Programming:	Python, MATLAB, C/C++, R, Batch Scripting
Python Libraries & Frameworks:	NumPy, Pandas, multiprocessing, threading, Tkinter, wxPython, Python.NET, real-time data acquisition, hardware integration, GUI development, CSV/data processing
Data Acquisition:	HBM QuantumX, Quanser, dSPACE, NI DAQ
Engineering Software:	MATLAB & Simulink, LabVIEW, ControlDesk, ETAS (INCA), Mathematica, GMSim, AutoVal, SOLIDWORKS, CATIA
Technical:	Real-time Systems, GUI Development (PyQt/Tkinter), Machine Learning (Neural Networks), Signal Processing, TCP/IP Communication, System Validation
Development Tools:	Eclipse, Visual Studio, Git
General:	Windows/Mac OS, MS Office, LaTeX
Additional exposure:	Linux, PLC programming